

## 3 Configuration

### 3.1 Overview

#### Introduction

Configuration is a dedicated operation where you use straightforward keystroke sequences to select and establish (configure) pertinent control data best suited for your application.

To assist you in the configuration process, there are prompts that appear in the upper and lower displays. These prompts let you know what group of configuration data (Set Up prompts) you are working with and also, the specific parameters (Function prompts) associated with each group.

Table 3-1 shows an overview of the prompt hierarchy as it appears in the controller.

#### What's in this section?

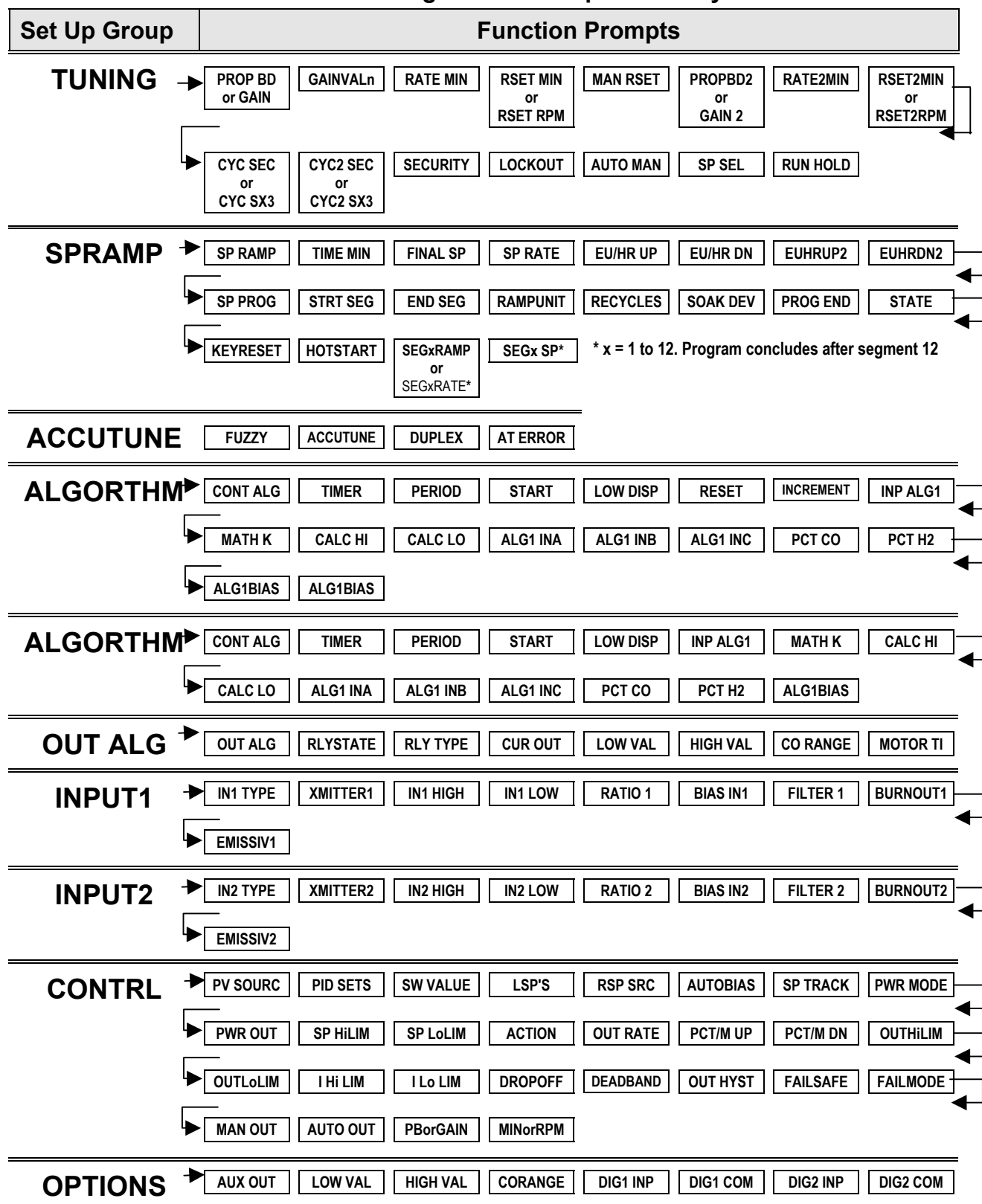
The following topics are covered in this section.

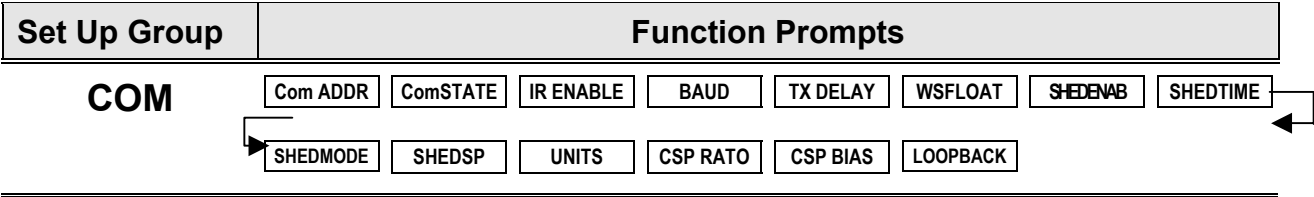
**Table 3-1 Configuration Topics**

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## 3.2 Configuration Prompt Hierarchy

Table 3-2 Configuration Prompt Hierarchy





| Set Up Group | Function Prompts |                            |          |          |          |           |          |          |         |   |
|--------------|------------------|----------------------------|----------|----------|----------|-----------|----------|----------|---------|---|
| ALARMS       | ➔                | A1S1TYPE                   | A1S1 VAL | A1S1 H L | A1S1 EV  | A1S2 TYPE | A1S2 VAL | A1S2 H L | A1S2 EV | ⬅ |
|              | ➔                | A2S1TYPE                   | A2S1 VAL | A2S1 H L | A2S1 EV  | A2S2TYPE  | A2S2 VAL | A2S2 H L | A2S2 EV | ⬅ |
|              | ➔                | AL HYST                    | ALM OUT1 | BLOCK    | DIAGNOST |           |          |          |         |   |
| DISPLAY      | ➔                | DECIMAL                    | TEMPUNIT | PWR FREQ | RATIO 2  | LANGUAGE  |          |          |         |   |
| CALIB        | ➔                | USED FOR FIELD CALIBRATION |          |          |          |           |          |          |         |   |
| STATUS       | ➔                | VERSION                    | FAILSAFE | TESTS    |          |           |          |          |         |   |

### 3.3 Configuration Procedure

#### Introduction

Each of the Set Up groups and their functions are pre-configured at the factory. The factory settings are shown in Table 3-4 through Table 3-14 that follow this procedure.

If you want to change any of these selections or values, follow the procedure in Table 3-3. This procedure tells you the keys to press to get to any Set Up group and any associated Function parameter prompt.

#### Procedure

##### ATTENTION

The prompting scrolls at a rate of 2/3 seconds when the **SET UP** or **FUNCTION** key is held in. Also,  or  keys will move group prompts forward or backward at a rate twice as fast.

**Table 3-3 Configuration Procedure**

| Step | Operation                     | Press                                                                                                                                                                      | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Enter Set Up Mode             |                                                                                           | <i>Upper Display</i> = <b>SETUP</b><br><i>Lower Display</i> = <b>TUNING</b> (This is the first Set Up Group title)                                                                                                                                                                                                                                                                                                                                                                                           |
| 2    | Select any Set Up Group       |                                                                                           | Sequentially displays the other Set Up group titles shown in the prompt hierarchy in Table 3-2 Configuration Prompt Hierarchy. You can also use the  or  keys to scan the Set Up groups in both directions. Stop at the Set Up group title that describes the group of parameters you want to configure. Then proceed to the next step. |
| 3    | Select a Function Parameter   |                                                                                         | <i>Upper Display</i> = the current value or selection for the first function prompt of the selected Set Up group.<br><i>Lower Display</i> = the first Function prompt within that Set Up group. Sequentially displays the other function prompts of the Set Up group you have selected. Stop at the function prompt that you want to change, then proceed to the next step.                                                                                                                                  |
| 4    | Change the Value or Selection |  or  | Increments or decrements the value or selection that appears for the selected function prompt. If you change the value or selection of a parameter while in Set Up mode but then decide not to enter it, press the <b>MAN/AUTO</b> key once. This will recall the original configuration. This "recall" procedure does not work for a Field Calibration process. Field Calibration is a one-way operation.                                                                                                   |
| 5    | Enter the Value or Selection  |                                                                                         | Enters value or selection made into memory after another key is pressed.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6    | Exit Configuration            |                                                                                         | Exits configuration mode and returns controller to the same state it was in immediately preceding entry into the Set Up mode. It stores any changes you have made.<br>If you do not press any keys for 30 seconds, the controller times out and reverts to the mode and associated display used prior to entry into Set Up mode.                                                                                                                                                                             |

## 3.4 Tuning Set Up Group

### Introduction

Tuning consists of establishing the appropriate values for the tuning constants you are using so that your controller responds correctly to changes in process variable and setpoint. You can start with predetermined values but you will have to watch the system to see how to modify them. The Accutune feature automatically selects Gain, Rate, and Reset on demand.

#### ATTENTION

Because this group contains functions that have to do with security and lockout, we recommend that you configure this group last, after all other configuration data has been loaded.

### Function Prompts

**Table 3-4 TUNING Group Function Prompts**

| Function Prompt<br>Lower Display    | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PROP BD</b><br>or<br><b>GAIN</b> | 0.1 to 9999 %<br>or<br>0.001 to 1000               | <p><b>PROPORTIONAL BAND</b> (simplex) is the percent of the range of the measured variable for which a proportional controller will produce a 100 % change in its output.</p> <p><b>GAIN</b> is the ratio of output change (%) over the measured variable change (%) that caused it.</p> $G = \frac{100\%}{PB\%}$ <p>where PB is the proportional band (in %)</p> <p>If the PB is 20 %, then the Gain is 5. And, at those settings, a 3 % change in the error signal (SP-PV) will result in a 15 % change in the controller's output due to proportional action. If the Gain is 2, then the PB is 50 %.</p> <p>Also defined as "HEAT" Gain on Duplex models for variations of Heat/Cool applications.</p> <p><i>The selection of Proportional Band or Gain is made in the CONTROL parameter group under prompt PBoRGAIn.</i></p> |
| <b>RATE MIN</b>                     | 0.00 to 10.00 minutes                              | <p><b>RATE</b> action, in minutes, affects the controller's output whenever the deviation is changing; and affects it more when the deviation is changing faster.</p> <p>Also defined as "HEAT" Rate on Duplex models for variations of Heat/Cool applications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Function Prompt<br>Lower Display         | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RSET MIN</b><br>or<br><b>RSET RPM</b> | 0.02 to 50.00                                      | <p><b>RSET MIN</b> = Reset in Minutes per Repeat<br/> <b>RSET RPM</b> = Reset in Repeats per Minute</p> <p><b>RESET</b> (or <b>Integral Time</b>) adjusts the controller's output in accordance with both the size of the deviation (SP–PV) and the time that it lasts. The amount of the corrective action depends on the value of Gain. The Reset adjustment is measured as how many times proportional action is repeated per minute or how many minutes before one repeat of the proportional action occurs.</p> <p>Used with control algorithm PID-A or PID-B.<br/> Also defined as "HEAT" Reset on Duplex models for variations of Heat/Cool applications.</p> <p><b>ATTENTION</b> The selection of whether Minutes per Repeat or Repeats per Minute is used is made in the CONTROL parameters group under the prompt <b>MINorRPM</b>.</p> |
| <b>MAN RSET</b>                          | –100 to +100<br>(in % output)                      | <p><b>MANUAL RESET</b> is only applicable if you use control algorithm PD WITH MANUAL RESET in the Algorithm Set Up group. Because a proportional controller will not necessarily line out at setpoint, there will be a deviation (offset) from setpoint. This eliminates the offset and lets the PV line out at setpoint.</p> <p><b>ATTENTION</b> Bias is shown on the lower display.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>PROBD2</b><br>or<br><b>GAIN 2</b>     | 0.1 to 9999 %<br>or<br>0.001 to 1000               | <p><b>PROPORTIONAL BAND 2</b> or <b>GAIN 2</b>, <b>RATE 2</b>, and <b>RESET 2</b> parameters are the same as previously described for "Heat" except that they refer to the <b>cool</b> zone tuning constants on duplex models or the second set of PID constants, whichever is pertinent.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>RATE2MIN</b>                          | 0.00 to 10.00 minutes                              | <p>This is the same as above except that it applies to Duplex models for the "COOL" zone of Heat/Cool applications or for the second set of PID constants.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>RSET2MIN</b><br><b>RSET2RPM</b>       | 0.02 to 50.00                                      | <p>These are the same as above except that they apply to Duplex models for the "COOL" zone of Heat/Cool applications or for the second set of PID constants.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Function Prompt<br>Lower Display         | Selections or<br>Range of Setting<br>Upper Display   | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CYC SEC</b><br>or<br><b>CYC SX3</b>   | 1 to 120                                             | <p><b>CYCLE TIME (HEAT)</b> determines the length of one time proportional output relay cycle. Defined as "HEAT" cycle time for Heat/Cool applications.</p> <p><b>CYC SEC</b>—Electromechanical relays<br/> <b>CYC SX3</b>—Solid state relays</p> <p><b>ATTENTION</b> <i>Cycle times are in either second or 1/3-second increments depending upon the configuration of RLY TYPE in the Output Algorithm Set Up group.</i></p>                                                                                                                                                                                                           |
| <b>CYC2 SEC</b><br>or<br><b>CYC2 SX3</b> | 1 to 120                                             | <p><b>CYCLE TIME 2 (COOL)</b> is the same as above except it applies to Duplex models as the cycle time in the "COOL" zone of Heat/Cool applications or for the second set of PID constants.</p> <p><b>CYC2 SEC</b>—Electromechanical relays<br/> <b>CYC2 SX3</b>—Solid state relays</p> <p><b>ATTENTION</b> <i>Cycle times are in either second or 1/3-second increments depending upon the configuration of RLY TYPE in the Output Algorithm Set Up group.</i></p>                                                                                                                                                                    |
| <b>SECURITY</b>                          | 0 to 9999                                            | <p><b>SECURITY CODE</b>—The level of keyboard lockout may be changed in the Set Up mode. Knowledge of a security code may be required to change from one level to another. This configuration should be copied and kept in a secure location.</p> <p>NOTE: The Security Code is for keyboard entry only and is not available via communications.</p> <p><b>ATTENTION</b> <i>Can only be changed if LOCKOUT selection is NONE.</i></p>                                                                                                                                                                                                   |
| <b>LOCKOUT</b>                           | <p>NONE</p> <p>CALIB</p> <p>+ CONF</p> <p>+ VIEW</p> | <p><b>LOCKOUT</b> applies to one of the functional groups: Configuration, Calibration, Tuning, Accutune. <b>DO NOT CONFIGURE UNTIL ALL CONFIGURATION IS COMPLETE.</b></p> <p><b>NONE</b>—No lockout; all groups are read/write.</p> <p><b>CALIB</b>—All groups are available for read/write except for the Calibration and Keyboard Lockout groups.</p> <p><b>+ CONF</b>—Tuning, SP Ramp, and Accutune groups are read/write. All other groups are read only. Calibration and Keyboard Lockout groups are not available.</p> <p><b>+ VIEW</b>—Tuning and Setpoint Ramp parameters are read/write. No other parameters are viewable.</p> |



| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                 |
|----------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | MAX                                                | <b>MAX</b> —Tuning and Setpoint Ramp parameters are available for read only. No other parameters are viewable.                                                                                                                                                                                                          |
| <b>AUTO MAN</b>                  | DISABLE<br>ENABLE                                  | <b>MANUAL/AUTO KEY LOCKOUT</b> —Allows you to disable the Manual/Auto key<br><br><b>DISABLE</b><br><b>ENABLE</b><br><br><b>ATTENTION</b> <i>Can only be viewed if LOCKOUT is configured for NONE.</i>                                                                                                                   |
| <b>RUN HOLD</b>                  | DISABLE<br>ENABLE                                  | <b>RUN/HOLD KEY LOCKOUT</b> —Allows you to disable the Run/Hold key, for either SP Ramp or SP Program. <i>The Run/Hold key is never disabled when used to acknowledge a latched alarm 1</i><br><br><b>DISABLE</b><br><b>ENABLE</b><br><br><b>ATTENTION</b> <i>Can only be viewed if LOCKOUT is configured for NONE.</i> |
| <b>SP SEL</b>                    | DISABLE<br>ENABLE                                  | <b>SETPOINT SELECT KEY LOCKOUT</b> —Allows you to disable the Setpoint Select key<br><br><b>DISABLE</b><br><b>ENABLE</b><br><br><b>ATTENTION</b> <i>Can only be viewed if LOCKOUT is configured for NONE.</i>                                                                                                           |

## 3.5 SP Ramp Set Up Group

### Introduction

Set Point Ramp, Set Point Programs and Set Point Rates can be configured in this group.

A single *Setpoint Ramp* [SP RAMP] can be configured to occur between the current local setpoint and a final local setpoint over a time interval of from 1 to 255 minutes.

A *Set Point Rate* [SPRATE] lets you configure a *specific rate of change* for any local setpoint change.

A single *Set Point Program* [SP PROG] with up to 12 segments can be configured.

For more information on Set Point Rate, Ramp and Programming, see Sections 4.17 through 4.20.

You can start and stop the ramp/program using the **RUN/HOLD** key.

*PV Hot Start* is a configurable feature and means that, at initialization, the setpoint is set to the current PV value and the Ramp or Rate or Program then starts from this value.

### Function Prompts

**Table 3-5 SPRAMP Group Function Prompts**

| Function Prompt<br>Lower Display                                                       | Selections or<br>Range of Setting<br>Upper Display            | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SP RAMP</b><br><br><i>SP Program must be disabled for SP Ramp prompts to appear</i> | <br><br><br><br><br><br><br><br>DISABLE<br><br><br><br>ENABLE | <b>SINGLE SETPOINT RAMP</b> —Make a selection to enable or disable the setpoint ramp function. Make sure you configure a ramp time and a final setpoint value.<br><br><i>SP Programming must be disabled.</i><br><br><b>DISABLE SETPOINT RAMP</b> —Disables the setpoint ramp option.<br><br><b>ENABLE SETPOINT RAMP</b> —Allows the single setpoint ramp prompts to be shown. |
| <b>TIME MIN</b>                                                                        | 0 to 255 minutes                                              | <b>SETPOINT RAMP TIME</b> —Enter the number of minutes desired to reach the final setpoint. A ramp time of “0” implies an immediate change of setpoint.                                                                                                                                                                                                                        |

| Function Prompt<br>Lower Display                                                                                                                               | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FINAL SP</b>                                                                                                                                                | Within setpoint limits                             | <p><b>SETPOINT RAMP FINAL SETPOINT</b>—Enter the value desired for the final setpoint. The controller will operate at the setpoint set here when ramp is ended.</p> <p><b>ATTENTION</b> If the ramp is on HOLD, the held setpoint can be changed by the ▲ and ▼ keys. However, the ramp time remaining and original ramp rate is not changed. Therefore, when returning to RUN mode, the setpoint will ramp at the same rate as previous to the local setpoint change and will stop if the final setpoint is reached before the time expires. If the time expires before the final setpoint is reached, it will jump to the final setpoint.</p> <p><b>ATTENTION</b> SP RAMP and SP RATE will cause the SP portion of Accutune to abort. PV Tune will continue to function normally. Ramp is placed into HOLD while tuning (TUNE configuration).</p> |
| <b>HOTSTART</b>                                                                                                                                                | DISABLE<br>ENABLE                                  | <p><b>DISABLE</b>—LSP1 is used as the initial ramp setpoint.</p> <p><b>ENABLE</b>—Current PV value is used as the initial ramp setpoint.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>SP RATE</b><br><br><i>SP Rate operates only when both SP Ramp and SP Programming are in HOLD mode or when both SP Ramp and SP Programming are disabled.</i> | DISABLE<br><br>ENABLE                              | <p><b>SETPOINT RATE</b>—Lets you configure a specific rate of change for any local setpoint change.</p> <p><b>DISABLE SETPOINT RATE</b>—Disables the setpoint rate option.</p> <p><b>ENABLE SETPOINT RATE</b>—Allows the SP rate feature.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>EU/HR UP</b>                                                                                                                                                | 0 to 9999 in engineering units per hour            | <p><b>RATE UP</b>—Rate up value. When making a setpoint change, this is the rate at which the controller will change from the original setpoint up to the new one. The ramping (current) setpoint can be viewed as SPn in the lower display.</p> <p>Entering a 0 will imply an immediate step change in Setpoint (i.e., no rate applies).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>EU/HR DN</b>                                                                                                                                                | 0 to 9999 in engineering units per hour            | <p><b>RATE DOWN</b>—Rate down value. When making a setpoint change, this is the rate at which the controller will change from the original setpoint down to the new one. The ramping (current) setpoint can be viewed as SPn in the lower display.</p> <p>Entering a 0 will imply an immediate step change in Setpoint (i.e., no rate applies).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Function Prompt<br>Lower Display                                                                                                                                                         | Selections or<br>Range of Setting<br>Upper Display                                                                                       | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SP PROG</b><br>(optional feature)<br><br><i>SP Ramp must be disabled for SP Program prompts to appear. If SP Rate is enabled, it does not operate while an SP Program is running.</i> | DISABLE<br>ENABLE                                                                                                                        | <b>SETPOINT RAMP/SOAK PROGRAM</b> —Available only with controllers that contain this option.<br><br><i>SP RAMP must be disabled.</i><br><br><b>DISABLE</b> —Disables setpoint programming.<br><b>ENABLE</b> —Enables setpoint programming.<br><br><b>ATTENTION</b> Detailed information for the prompts for SP Programming may be found <i>Section 4.17</i> The listing below is only for reference purposes. |
| <b>STRT SEG</b>                                                                                                                                                                          | 1 to 11                                                                                                                                  | Start Segment Number                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>END SEG</b>                                                                                                                                                                           | 2 to 12 even numbers<br>Always end in a soak segment<br>(2, 4, ... 12)                                                                   | End Segment Number                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>RAMPUNIT</b>                                                                                                                                                                          | TIME<br>EU/MIN<br>EU/HR                                                                                                                  | <b>RAMPUNIT</b> —Engineering Units for Ramp Segments<br><br><b>TIME</b> in hours: minutes<br><b>RATE</b> in Engineering units per minute<br><b>RATE</b> in Engineering units per hour                                                                                                                                                                                                                         |
| <b>RECYCLES</b>                                                                                                                                                                          | 0 to 99 recycles                                                                                                                         | Number of Program Recycles                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>SOAK DEV</b>                                                                                                                                                                          | 0 to 99<br>The number selected will be the PV value (in engineering units) above or below the setpoint outside of which the timer halts. | Guaranteed Soak Deviation Value                                                                                                                                                                                                                                                                                                                                                                               |
| <b>PROG END</b>                                                                                                                                                                          | LASTSP (Hold at last setpoint in the program)<br>F SAFE (Manual mode/Failsafe output)                                                    | Program Termination State                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>STATE</b>                                                                                                                                                                             | DISABLE<br>HOLD                                                                                                                          | Program State at Program End                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>KEYRESET</b>                                                                                                                                                                          | DISABLE<br>ToBEGIN<br>RERUN                                                                                                              | Reset/Rerun SP Program                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>HOTSTART</b>                                                                                                                                                                          | DISABLE<br>ENABLE                                                                                                                        | Hot Start                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>SEG1RAMP</b> or <b>SEG1RATE</b>                                                                                                                                                       | 0-99 hours.0-59 minutes<br>Engineering units/minute<br>or<br>Engineering units/hour                                                      | Segment #1 Ramp Time or Segment #1 Ramp Rate<br><br><b>Select TIME, EU/MIN, or EU/HR at prompt RAMPUNIT. All ramps will use the same selection.</b>                                                                                                                                                                                                                                                           |

| Function Prompt<br>Lower Display                                                                                                                                                                                                                                                                                                                                           | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------|
| <b>SEG2 SP</b>                                                                                                                                                                                                                                                                                                                                                             | Within the Setpoint limits                         | Segment #2 Soak Setpoint Value |
| <b>SEG2TIME</b>                                                                                                                                                                                                                                                                                                                                                            | 0-99 hours.0-59 minutes                            | Segment #2 Soak Duration       |
| <b>SEG3RAMP or<br/>SEG3RATE</b><br><br><b>SEG4 SP<br/>SEG4TIME</b><br><br><b>SEG5RAMP or<br/>SEG5RATE</b><br><br><b>SEG6 SP<br/>SEG6TIME</b><br><br><b>SEG7RAMP or<br/>SEG7RATE</b><br><br><b>SEG8 SP<br/>SEG8TIME</b><br><br><b>SEG9RAMP or<br/>SEG9RATE</b><br><br><b>SG10 SP<br/>SG10TIME</b><br><br><b>SG11RAMP or<br/>SG11RATE</b><br><br><b>SG12 SP<br/>SG12TIME</b> | Selections are same as above.                      | Same as above                  |

## 3.6 Accutune Set Up Group

### Introduction

*Accutune III* automatically calculates GAIN, RATE, and RESET TIME (PID) tuning constants for your control loop. When initiated on demand, the Accutune algorithm measures a process step response and automatically generates the PID tuning constants needed for no overshoot on your process.

*Fuzzy*, Fuzzy Overshoot Suppression: When enabled, this configuration will suppress or eliminate any overshoot that may occur as a result of the existing tuning parameters, as the PV approaches the setpoint.

*Tune*, Demand Tuning: The tuning process is initiated through the operator interface keys or via a digital input (if configured). The algorithm then calculates new tuning parameters and enters them in the tuning group. *Tune* will operate with PIDA, PIDB, PD+MR and Three Position Step Control algorithms.

*SP*, SP Tuning: SP tuning continuously adjusts the PID parameters in response to setpoint changes. You can select tuning on minimum setpoint changes of 5 % up to 15 % span. Perform SP tuning after you have configured the controller. SP Tuning does not operate with the Three Position Step Control algorithm.

*Simplex* Tuning is used when a Simplex Control Algorithm is configured and uses the current SP value and alters the output over the Output Limit Range.

*Duplex* Tuning is used when a Duplex Control Algorithm is configured. To perform a Duplex Tune, Two Local Setpoints must be configured per the Control Group in Section 3.11.

### Function Prompts

**Table 3-6 ACCUTUNE Group Function Prompts**

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FUZZY</b>                     | DISABLE<br><br>ENABLE                              | <p><b>FUZZY OVERSHOOT SUPPRESSION</b>—Can be enabled or disabled independently of whether Demand Tuning or SP Tuning is enabled or disabled.</p> <p><b>DISABLE</b>—Disables Fuzzy Overshoot Suppression.</p> <p><b>ENABLE</b>—The instrument uses Fuzzy Logic to suppress or minimize any overshoot that may occur when PV approaches SP. It will not recalculate any new tuning parameters.</p> |

| Function Prompt<br>Lower Display                                                                         | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ACCUTUNE</b>                                                                                          | DISABLE<br><br>TUNE                                | <b>ACCUTUNE III</b><br><br><b>DISABLE</b> —Disables the Accutune function.<br><br><b>DEMAND TUNING</b> —If TUNE is selected, and tuning is initiated through the operator interface or digital input (if configured), the algorithm calculates new tuning parameters and enters them into the tuning group. This tuning requires no process knowledge and does not require line out for initialization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>DUPLEX</b><br><br><i>This prompt only appears when a Duplex Control Algorithm has been configured</i> | MANUAL<br><br>AUTO<br><br>DISABLE                  | <b>DUPLEX ACCUTUNING III</b> – These prompts only appear when a duplex output type has been configured.<br><br><b>MANUAL</b> – Tune manually using LSP 1 and LSP 2 values. LSP 1 is used to derive tuning parameters associated with HEAT (output > 50 %). LSP 2 is used to derive tuning parameters associated with COOL (output < 50 %).<br><br><b>AUTOMATIC</b> – Tuning is performed automatically on both HEAT and COOL sequentially. LSP 1 is used for HEAT tuning and LSP 2 is used for COOL tuning. To initiate tuning, either LSP 1 or LSP 2 must be in use.<br><br><b>DISABLE</b> – The current SetPoint is used to derive a single set of blended tuning parameters. This tuning is performed over the range of the output limits similar to Simplex Tuning. The Tuning Parameters derived are placed into both the HEAT and COOL tune sets (PID 1 and PID 2). |
| <b>AT ERROR</b><br>(Read Only)                                                                           | NONE<br><br>RUNNING<br><br>ABORT<br><br>SP2        | <b>ACCUTUNE ERROR STATUS</b> —When an error is detected in the Accutune process, an error prompt will appear.<br><br><b>NONE</b> —No errors occurred during last Accutune procedure.<br><br><b>RUNNING</b> —An Accutune process is still active checking process gain, even though “T” is not lit. It does not affect keyboard operation.<br><br><b>CURRENT ACCUTUNE PROCESS ABORTED</b> —Caused by one of the following conditions:<br><ul style="list-style-type: none"> <li>• changing to manual mode</li> <li>• digital input detected</li> <li>• in heat region of output but a cool output was calculated, or vice versa.</li> </ul><br><b>SP2</b> —LSP2 not configured or a Setpoint other than LSP1 or LSP2 is in use.                                                                                                                                            |

## 3.7 Algorithm Set Up Group

### Introduction

This data deals with various algorithms in the controller and Timer functions.

The Timer section allows you to configure a time-out period and to select the timer start by either the keyboard (**RUN/HOLD** key) or Alarm 2. An optional digital input can also be configured to the start the timer. The timer display is selectable as either “time remaining” (*see TI REM*) or “elapsed time” (*see E TIME*).

Alarm 1 is activated at the end of the time-out period. When the timer is enabled, it has exclusive control of the alarm 1 relay—any previous alarm 1 configuration is ignored. At time-out, the timer is ready to be activated again by whatever action has been configured.

### Function Prompts

**Table 3-7 ALGORITHM Group Function Prompts**

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CONT ALG</b>                  | ON-OFF                                             | <p>The <b>CONTROL ALGORITHM</b> lets you select the type of control that is best for your process.</p> <p><b>ON/OFF</b> is the simplest control type. The output can be either ON (100 %) or OFF (0 %). The Process Variable (PV) is compared with the setpoint (SP) to determine the sign of the error (<math>ERROR = PV - SP</math>). The ON/OFF algorithm operates on the sign of the error signal.</p> <p>In Direct Acting Control, when the error signal is positive, the output is 100 %; and when the error signal is negative, the output is 0 %. If the control action is reverse, the opposite is true. An adjustable overlap (Hysteresis Band) is provided between the on and off states.</p> <p><b>ATTENTION</b> Other prompts affected: <i>OUT HYST</i></p> <p><b>DUPLEX ON/OFF</b> is an extension of this algorithm when the output is configured for a Duplex control algorithm. It allows the operation of a second ON/OFF output. There is a deadband between the operating ranges of the two inputs and an adjustable overlap (hysteresis) of the on and off states of each output. Both Deadband and Hysteresis are separately adjustable. With no relay action the controller will read 50 %.</p> <p><b>ATTENTION</b> Other prompts affected: <i>OUT HYST and DEADBAND</i></p> |



| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display                                                                                                            | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | PID A<br><br><b>ATTENTION</b> PID A should not be used for Proportional only action; i.e., no integral (reset) action. Instead, use PD+MR with rate set to 0. | <p><b>PID A</b> is normally used for three-mode control. This means that the output can be adjusted somewhere between 100 % and 0 %. It applies all three control actions—Proportional (P), Integral (I), and Derivative (D)—to the error signal.</p> <p><b>Proportional (Gain)</b>—Regulates the controller's output in proportion to the error signal (the difference between Process Variable and Setpoint).</p> <p><b>Integral (Reset)</b>—Regulates the controller's output to the size of the error and the time the error has existed. (The amount of corrective action depends on the value of proportional Gain.)</p> <p><b>Derivative (Rate)</b>—Regulates the controller's output in proportion to the rate of change of the error. (The amount of corrective action depends on the value of proportional Gain.)</p> |
|                                  | PID B                                                                                                                                                         | <p><b>PID B</b>—Unlike the PID A equation, the controller gives only an integral response to a setpoint change, with no effect on the output due to the gain or rate action, and it gives full response to PV changes. Otherwise controller action is as described for the PID A equation. See note on PID A.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                  | PD+MR                                                                                                                                                         | <p><b>PD WITH MANUAL RESET</b> is used whenever integral action is not wanted for automatic control. The equation is computed with no integral contribution. The MANUAL RESET, which is operator adjustable, is then added to the present output to form the controller output.</p> <p>Switching between manual and automatic mode will be bumpless.</p> <p>If you select PD with Manual Reset you can also configure the following variations:</p> <ul style="list-style-type: none"> <li>• PD (Two Mode) control,</li> <li>• P (Single Mode) control.</li> </ul> <p>Set Rate (D) to 0.</p> <p><b>ATTENTION</b> Other prompts affected: MAN RSET in the Tuning Set Up group</p>                                                                                                                                                |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CONT ALG</b><br>(continued)   | 3PSTEP                                             | <p><b>THREE POSITION STEP</b>—The Three Position Step Control algorithm allows the control of a valve (or other actuator) with an electric motor driven by two controller relay outputs; one to move the motor upscale, the other downscale without a feedback slidewire linked to the motor shaft. The deadband is adjustable in the same manner as the duplex output algorithm.</p> <p>The Three Position Step Control algorithm provides an output display (OUT) which is an estimated motor position, since the motor is not using any slidewire feedback. Although this output indication is only an approximation, it is “corrected” each time the controller drives the motor to one of its stops (0 % or 100 %). It avoids all the control problems associated with the feedback slidewire (wear, dirt, noise). When operating in this algorithm, the estimated OUT display is shown to the nearest percent (i.e., no decimal). This selection forces the Output Algorithm selection to “POSITON”. See <i>Subsection 3.8</i>.</p> <p>Refer to the <i>Operation</i> section for motor position displays.</p> <p>As a customer configurable option, when a second input board is installed, the motor slidewire can be connected to the controller. The actual slidewire position is then shown on the lower display as POS. <b>This value is used for display only. It is NOT used in the Three Position Step algorithm.</b> To configure this option, set Input 2 actuation to SLIDEW and then calibrate Input 2 per <i>Subsection 6.4</i>.</p> <p><b>ATTENTION</b> Other prompts affected: DEADBAND</p> |
| <b>TIMER</b>                     | DISABLE<br>ENABLE                                  | <p><b>TIMER</b> allows you to enable or disable the timer option.</p> <p>The timer option allows you to configure a timeout period and to select timer start by either the keyboard (RUN/HOLD key) or Alarm 2. A digital input can also be configured to start the timer.</p> <p>When the timer is enabled, it has exclusive control of the alarm 1 relay; any previous alarm configuration is ignored. At timeout, the timer is ready to be re-activated by whatever action has been configured. Alarm 1 is activated at the end of the timeout period.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>PERIOD</b>                    | 0:00 to 99:59                                      | <b>PERIOD</b> allows you to configure the length of timeout period (from 0 to 99 hours: 59 minutes).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>START</b>                     | KEY<br>ALARM 2                                     | <b>START</b> allows you to select whether the timer starts with the keyboard (Run/Hold key) or Alarm 2.                                                                                                                                                                                                                                                                                                                                                                |
| <b>LOW DISP</b>                  | TI REM<br>E TIME                                   | <p><b>LOW DISP</b> allows you to select whether time remaining (TI REM) or elapsed time (E TIME) is displayed for the timer option.</p> <p>The time is shown on the lower display in HH:MM format along with a rotating “clock” character.</p> <ul style="list-style-type: none"> <li>• If the “clock” rotation is <i>clockwise</i>, elapsed time is indicated.</li> <li>• If the “clock” rotation is <i>counterclockwise</i>, time remaining is indicated.</li> </ul> |
| <b>RESET</b>                     | KEY<br>ALARM 1                                     | <b>RESET</b> allows you to reset the timer with the keyboard (Run/Hold key) or Alarm 1.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>INCREMENT</b>                 | MINUTES<br>SECONDS                                 | <b>INCREMENT</b> allows you to increment the timer in either minutes or seconds.                                                                                                                                                                                                                                                                                                                                                                                       |

**INPUT MATH ALGORITHMS**—Controllers with two inputs are provided with one input algorithm. Unless otherwise noted, these selections are provided only as part of the Math Options package. Each algorithm can be configured to provide a derived (calculated) PV or a derived Remote Setpoint. Up to three inputs may be applied to the calculation. See Inputs A, B, and C for definitions per equation.

All algorithms operate in engineering units except Feedforward (F FWRD) which operates in percent of output units.

**ATTENTION** When the Input C configuration is set to NONE, the value of Input C used in the functions is automatically set to 1.0, except for the Summer algorithm, where it is set to 0.0.

|                 |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INP ALG1</b> | NONE                                                                                                                                                                                                                                                            | <p><b>INPUT ALGORITHM 1</b> has the following selections from which to choose:</p> <p><b>NONE</b>—No algorithm configured</p>                                                                                                                                                                                                                                                              |
|                 | W AVG<br>(See Note 2)<br><br>(Standard feature on controllers with two analog inputs)<br>$\text{Alg1} = [(\text{Input A} \times \text{Ratio A} + \text{Bias A}) + (K \times \text{Input B} \times \text{Ratio B} + \text{Bias B})] / (1 + K) + \text{Alg1Bias}$ | <p><b>WEIGHTED AVERAGE</b>—When you configure for Weighted Average, the controller will compute a PV or SP for the control algorithm from the following equation:</p>                                                                                                                                                                                                                      |
|                 | F FWRD<br><br>(Standard feature on controllers with two analog inputs)                                                                                                                                                                                          | <p><b>FEEDFORWARD SUMMER</b>—Feedforward uses Input A, following a Ratio and Bias calculation as a value summed directly with the PID computed output value and sent, as an output value, to the final control element.</p> <p>This algorithm will only function in automatic mode and is not used for Three Position Step Control applications.</p> <p>The following formula applies:</p> |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display                                     | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | Controller Output = PID Output + (Input A x Ratio A + Bias A ) x (100 / Input A Range) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                  | FFWDMu<br>(Standard feature on controllers with two analog inputs)                     | <p><b>FEEDFORWARD MULTIPLIER</b>—Feedforward uses Input A, following a Ratio and Bias calculation as a value multiplied directly with the PID computed output value and sent, as an output value, to the final control element.</p> <p>This algorithm will only function in automatic mode and cannot be used for Three Position Step Control applications.</p> <p>The following formula applies:</p> $\text{Controller Output} = \text{PID Output} \times (\text{Input A} \times \text{Ratio A} + \text{Bias A}) / \text{Input A Range}$ |
|                                  | SUMMER<br>(See Note 2)                                                                 | <p><b>SUMMER WITH RATIO AND BIAS</b>—The following formula applies:</p> $\text{Alg1} = (\text{Input A} \times \text{Ratio A} + \text{Bias A}) + (\text{Input B} \times \text{Ratio B} + \text{Bias B}) + (\text{Input C} \times \text{Ratio C} + \text{Bias C}) + \text{Alg1Bias}$                                                                                                                                                                                                                                                        |
|                                  | HI SEL<br>(See Note 2)                                                                 | <p><b>INPUT HIGH SELECT WITH RATIO AND BIAS</b>—This selection specifies the PV or SP as the higher of Input 1 or Input 2. The following formula applies:</p> $\text{Alg1} = \text{higher of } (\text{Input A} \times \text{Ratio A} + \text{Bias A}) \text{ or } (\text{Input B} \times \text{Ratio B} + \text{Bias B})$                                                                                                                                                                                                                 |
|                                  | LO SEL<br>(See Note 2)                                                                 | <p><b>INPUT LOW SELECT WITH RATIO AND BIAS</b>—This selection specifies the PV or SP as the lower of Input 1 or Input 2. The following formula applies:</p> $\text{Alg1} = \text{lower of } (\text{Input A} \times \text{Ratio A} + \text{Bias A}) \text{ or } (\text{Input B} \times \text{Ratio B} + \text{Bias B})$                                                                                                                                                                                                                    |
|                                  | √MuDIV<br>(See Note 1)                                                                 | <p><b>MULTIPLIER DIVIDER WITH SQUARE ROOT</b>—The following formula applies:</p> $\text{Alg1} = K \times \text{Sq.Rt.} \{ (\text{Input A} \times \text{Ratio A} + \text{Bias A}) \times (\text{Input C} \times \text{Ratio C} + \text{Bias C}) / (\text{Input B} \times \text{Ratio B} + \text{Bias B}) \}$ $\times (\text{Calc Hi} - \text{Calc Lo}) + \text{Alg1Bias}$ <p><b>See Figure 3-1 at the end of this section for an example of Mass Flow Compensation using the Multiplier/Divider Algorithm.</b></p>                         |
|                                  | √MULT<br>(See Note 1)                                                                  | <p><b>MULTIPLIER WITH SQUARE ROOT</b>—The following formula applies:</p> $\text{Alg1} = K \times \text{Sq.Rt.} \{ (\text{Input A} \times \text{Ratio A} + \text{Bias A}) \times (\text{Input B} \times \text{Ratio B} + \text{Bias B}) \times (\text{Input C} \times \text{Ratio C} + \text{Bias C}) \}$ $\times (\text{Calc Hi} - \text{Calc Lo}) + \text{Alg1Bias}$                                                                                                                                                                     |
|                                  | MuDIV<br>(See Note 1)                                                                  | <p><b>MULTIPLIER DIVIDER</b>—The following formula applies:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Function Prompt<br>Lower Display                                                                                                                                                                                                                                                       | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Alg1} = K \times \left[ \frac{(\text{Input A} \times \text{Ratio A} + \text{Bias A}) \times (\text{Input C} \times \text{Ratio C} + \text{Bias C})}{(\text{Input B} \times \text{Ratio B} + \text{Bias B})} \right] \times (\text{Calc Hi} - \text{Calc Lo}) + \text{Alg1Bias}$ |                                                    |                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                        | MULT<br>(See Note 1)                               | <b>MULTIPLIER</b> —The following formula applies:<br><br>$\text{Alg1} = K \times [(\text{Input A} \times \text{Ratio A} + \text{Bias A}) \times (\text{Input C} \times \text{Ratio C} + \text{Bias C}) \times (\text{Input B} \times \text{Ratio B} + \text{Bias B})] \times (\text{Calc Hi} - \text{Calc Lo}) + \text{Alg1Bias}$                                      |
|                                                                                                                                                                                                                                                                                        | CARB A                                             | <b>CARBON POTENTIAL A</b> —Make this selection if you have a Cambridge or Marathon monitor type Zirconium Oxide sensor. See Note 3.                                                                                                                                                                                                                                    |
|                                                                                                                                                                                                                                                                                        | CARB B                                             | <b>CARBON POTENTIAL B</b> —Make this selection if you have a Corning type Zirconium Oxide sensor. This algorithm requires a temperature range within the region of 1380 to 2000°F. See Note 3.                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                        | CARB C                                             | <b>CARBON POTENTIAL C</b> —Make this selection if you have an A.A.C.C. type Zirconium Oxide sensor. This algorithm requires a temperature range within the region of 1380 °F to 2000 °F. See Note 3.                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                        | CARB D                                             | <b>CARBON POTENTIAL D</b> —Make this selection if you have a Barber Coleman, MacDhui, or Bricesco type Zirconium Oxide sensor. This algorithm requires a temperature range within the region of 1380 to 2000°F. See Note 3.                                                                                                                                            |
|                                                                                                                                                                                                                                                                                        | FCC                                                | <b>CARBON POTENTIAL FCC</b> —Make this selection if you have a Furnace Controls Corp Accucarb type Zirconium Oxide sensor. This algorithm requires a temperature range within the region of 1380 °F to 2000 °F. See Note 3.                                                                                                                                            |
|                                                                                                                                                                                                                                                                                        | DEW PT                                             | <b>DEWPOINT OF CARBONIZING ATMOSPHERE</b> —Use this selection if you are using any Zirconium Oxide Carbon Probe and you want to measure the atmosphere in terms of Dewpoint. The range is –50 °F to 100 °F or –48 °C to 38 °C. This algorithm requires a temperature range within the region of 1000 °F to 2200 °F and a minimum carbon probe value of 800 millivolts. |
|                                                                                                                                                                                                                                                                                        | OXYGEN                                             | <b>PERCENT OXYGEN RANGE</b> —Make this selection if you are using a Zirconium Oxide Oxygen Probe to measure Percent of Oxygen in a range of 0 to 40 % O <sub>2</sub> . This algorithm requires a temperature range within the region of 800 °F to 3000 °F.                                                                                                             |

| Function Prompt<br>Lower Display                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>ATTENTION</b> The Carbon and Dewpoint selections will automatically set the first input actuation to Carbon. The Oxygen selection will automatically set the first input actuation to Oxygen. Input 2 can be any input actuation, but it is normally a type K, R or S thermocouple input, depending upon the probe type selected. All calculations are performed by the Controller with Percent Carbon shown as the PV display. The actual value of each analog input may be viewed on the lower display. For all Carbon Types, if the value of Percent Carbon falls below 0.1% - such as can happen when the Carbon Probe voltage output falls below 900 mVdc – then the Controller will continue to update the PV display, but the accuracy is unspecified. Likewise, if the measured temperature falls outside of the specified ranges as noted above for the Carbon, Oxygen and Dewpoint input types, then the Controller will continue to update the PV display, but the accuracy is unspecified. For the Dewpoint algorithm, if the Carbon Sensor voltage falls below 800 mVdc, then the Dew Point is calculated as if the sensor voltage was at 800 mVdc.</p> |                                                    |                                                                                                                                                                                                                                  |
| <b>MATH K</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.001 to 1000 floating                             | <b>WEIGHTED AVERAGE RATIO OR MASS FLOW ORIFICE CONSTANT (K) FOR MATH SELECTIONS</b> —Only applicable for algorithms W AVG or General Math selections $\sqrt{\text{MuDIV}}$ , $\sqrt{\text{MULT}}$ , MuDIV, or MULT.              |
| <b>CALC HI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | –999. To 9999. Floating<br>(in engineering units)  | <b>CALCULATED VARIABLE HIGH SCALING FACTOR FOR INPUT ALGORITHM 1</b> —Used only when either Summer, Input Hi/Lo, or one of the General Math functions was selected as the Input Algorithm. See Note 2.                           |
| <b>CALC LO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | –999. To 9999. Floating<br>(in engineering units)  | <b>CALCULATED VARIABLE LOW SCALING FACTOR FOR INPUT ALGORITHM 1</b> —Used only when either Summer, Input Hi/Lo, or one of the General Math functions was selected as the Input Algorithm. See Note 2.                            |
| <b>ALG1 INA</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | INPUT 1<br>INPUT 2<br>OUTPUT                       | <b>ALGORITHM 1, INPUT A SELECTION</b> will represent one of the available selections.<br><br><b>Input 1</b><br><b>Input 2</b><br><b>Output</b> – Should not be used for Three Position Step Control applications)                |
| <b>ALG1 INB</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | INPUT 1<br>INPUT 2<br>OUTPUT                       | <b>ALGORITHM 1, INPUT B SELECTION</b> will represent one of the available selections.<br><br><b>Input 1</b><br><b>Input 2</b><br><b>Output</b> – Should not be used for Three Position Step Control applications)                |
| <b>ALG1 INC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NONE<br>INPUT 1<br>INPUT 2<br>OUTPUT               | <b>ALGORITHM 1, INPUT C SELECTION</b> will represent one of the available selections.<br><br><b>None</b><br><b>Input 1</b><br><b>Input 2</b><br><b>Output</b> – Should not be used for Three Position Step Control applications) |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                     |
|----------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PCT CO</b>                    | 0.020 to 0.350 (fractional percent of CO)          | <b>PERCENT CARBON</b> is only applicable when Carbon Potential is selected. Enter the value in percent carbon monoxide that is applicable for the enriching gas used in fractional form.<br><br>FOR EXAMPLE:<br>Natural Gas = 20.0 % CO, then setting is 0.200<br>Propane Gas = 23.0 % CO, setting is 0.230 |
| <b>ALG1 BIAS</b>                 | -999 to 9999 floating (in engineering units)       | <b>INPUT ALGORITHM 1 BIAS</b> —Does not apply to selections: FFWRD, FFWDMU, HISEL or LOSEL.                                                                                                                                                                                                                 |
| <b>PCT H2</b>                    | 1.0 to 99.0 (% H <sub>2</sub> )                    | <b>HYDROGEN CONTENT FOR DEWPOINT</b> is only applicable when Dewpoint is selected. Enter a value for the percentage of Hydrogen content that is applicable.                                                                                                                                                 |

**Math Algorithm Notes:**

1. Calculation ranges for the Math Algorithms are set via CALC HI and CALC LO parameters and are between -999. and 9999. The SP High and Low values (SP Range) are independent of these settings and can be any value between -999. and 9999.
2. The CALC HI and CALC LO values determine the range limits for the SP High and Low values for the Weighted Average, Summer, Hi Select and Low Select algorithms.
3. If the Ratio for Input 2 is set to 0.0, then a constant value may be used for the Input 2 value via the Input 2 Bias setting. For this configuration, the Input 2 low range and the Sooting diagnostic messages are disabled.

### Figure 3-1 Mass Flow Example

#### Example - Mass Flow Compensation

A gas flow rate of 650 SCFM develops a differential pressure of 90" H<sub>2</sub>O across an orifice plate at reference conditions of 30 psig and 140°F. Compensate this gas flow for temperature and pressure variations.

$$\text{Flow} = K \sqrt{\frac{DP_f \times P_f}{T_f} \times \frac{T_{ref}}{P_{ref}}}$$

Where:

f = flowing conditions

ref = reference conditions (in absolute units)

Apply Multiplier/Divider Algorithm:

$$PV = K \sqrt{\frac{(\text{Input A} \times \text{Ratio A} + \text{Bias A}) \times (\text{Input C} \times \text{Ratio C} + \text{Bias C})}{(\text{Input B} \times \text{Ratio B} + \text{Bias B})}} \times (\text{Calc HI} - \text{Calc LO})$$

Assign inputs using Engineering units:

Let:

Input A = DP<sub>f</sub> = IN1 (in H<sub>2</sub>O)

Input B = T<sub>f</sub> = IN2 + Bias2 = IN2°F + 460 (°R)

Input C = P<sub>f</sub> = IN3 + Bias3 = IN3psig + 14.7(psia)

T<sub>ref</sub> = 140°F + 460 = 600°R

P<sub>ref</sub> = 30 psig + 14.7 = 44.7 psia

Calc<sub>Hi</sub> = 650.0

Calc<sub>Lo</sub> = 0.0

Flow in SFCM at Reference Conditions

K = to be determined next

Note: If temperature and pressure signals are already ranged in absolute units, no Bias is required for inputs B and C.

$$PV = Q_{SCFM} = \sqrt{\frac{DP_f \times (IN3 + 14.7)}{(IN2 + 460)}} \times K^2 \times (650.0 - 0.0)$$

Note: When IN2 and IN3 are at the reference conditions of 600°R (140°F) and 44.7psia (30 psig) respectively and DP<sub>f</sub> = 90" H<sub>2</sub>O, the equation must calculate 650 SCFM. To accomplish this, divide the DP value by "90" to normalize the equation.

$$Q_{SCFM} = \sqrt{\frac{DP_f}{90} \times \frac{(IN3 + 14.7)}{(IN2 + 460)}} \times \frac{T_{ref}}{P_{ref}} \times 650$$

Rearranging terms:

$$Q_{SCFM} = \sqrt{\frac{DP_f \times \frac{(IN3 + 14.7)}{(IN2 + 460)}}{\frac{1}{90} \times \frac{T_{ref}}{P_{ref}}}} \times 650$$

Variable
Constant = K<sup>2</sup>

**Example continued  
on next page**

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**Example - Mass Flow Compensation - continued**

Determined value of K:

$$K^2 = \frac{1}{90} \times \frac{T_{\text{ref}}}{P_{\text{ref}}} = \frac{600}{(90)(44.7)} = 0.14914$$

Therefore K = 0.386

$$Q_{\text{SCFM}} = \underbrace{(0.386)}_{\boxed{K}} \underbrace{(650)}_{\boxed{(\text{Calc}_{\text{HI}} - \text{Calc}_{\text{LO}})}} \sqrt{\frac{DP_f \text{ (in H}_2\text{O)} (\text{IN3} + 14.7)}{(\text{IN2} + 460)}}$$

**Summary of Flow Values At Values Conditions**

|                         | Temp (T <sub>f</sub> )<br>(°R) | Pressure (T <sub>f</sub> )<br>(psia) | Flow (SFCM)                                  |                                               |
|-------------------------|--------------------------------|--------------------------------------|----------------------------------------------|-----------------------------------------------|
|                         |                                |                                      | DP <sub>f</sub> = 45" H <sub>2</sub> O (50%) | DP <sub>f</sub> = 90" H <sub>2</sub> O (100%) |
| Reference<br>Conditions | 140°F + 460                    | 30 psi + 14.7                        | 459                                          | 650                                           |
|                         | 170°F + 460                    | 50 psi + 14.7                        | 539                                          | 763                                           |
|                         | 170°F + 460                    | 20 psi + 14.7                        | 395                                          | 559                                           |
|                         | 110°F + 460                    | 50 psi + 14.7                        | 567                                          | 802                                           |
|                         | 110°F + 460                    | 20 psi + 14.7                        | 415                                          | 587                                           |

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## 3.8 Output Set Up Group

### Introduction

This group deals with various output types in the controller, the Digital Output Status and the Current Output operation.

**ATTENTION** The Tuning Group is automatically configured to have two PID sets when a Duplex Control Algorithm is selected.

### Function Prompts

**Table 3-8 OUTPUT Group Function Prompts**

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OUT ALG</b>                   |                                                    | <p>The <b>OUTPUT ALGORITHM</b> lets you select the type of output you want. <b>Not applicable with Control algorithm prompt 3PSTEP.</b></p> <p>Selections are hardware dependent. For example, if the controller does not have a current output, then none of the prompts for Output Algorithms that need a current output will appear. Likewise, if the controller does not have a relay output, then none of the prompts that need a relay output will appear.</p> <p><b>ATTENTION</b> For all Duplex Output forms, PID heat parameters apply for controller output greater than 50 %; PID cool parameters apply for controller output less than 50 %.</p> |
|                                  | TIME                                               | <p><b>TIME SIMPLEX</b>—This output algorithm uses Digital Output 1 for Time Proportional Control. The output is updated per the Loop sampling rate selection. Time Proportional Output has a resolution of 4.44 msec. Cycle Time is adjustable from 1 to 120 seconds.</p>                                                                                                                                                                                                                                                                                                                                                                                    |
|                                  | CURRENT                                            | <p><b>CURRENT SIMPLEX</b>—Type of output using one 4 mA to 20 mA signal that can be fed into a positive or negative grounded load of 0 to 1000 ohms. This signal can easily be configured for 4-20 mA or 0-20 mA operation via the CO RANGE configuration, below.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | POSITON                                            | <p><b>POSITION PROPORTIONAL</b> —Type of output using two relays and a motor that has a 100 to 1000 ohms feedback slidewire.</p> <p>This output algorithm selection forces Input 2 to the SLIDEW selection when the Control Algorithm is any selection other than 3PSTEP.</p> <p><b>ATTENTION</b> Other prompts affected: DEADBAND, IN2 TYPE</p>                                                                                                                                                                                                                                                                                                             |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | TIME D                                             | <b>TIME DUPLEX</b> —This output algorithm uses Digital Outputs 1 and 2 for Duplex Time Proportional Control. The outputs are updated per the Loop sampling rate selection. Time Proportional Output has a resolution of 4.44 msec. Cycle Time is adjustable from 1 second to 120 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                  | CUR D                                              | <b>CURRENT DUPLEX</b> is similar to current simplex but uses a second current output. The second output is usually scaled so that zero and span correspond with 0 % and 50 % output (cool zone). When the output is 0 % to 50 %, the controller uses tuning parameter set #2, when the output is 50 % to 100 % it uses set #1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                  | CUR TI                                             | <b>ATTENTION</b> Other prompts affected: OUT RNG<br><b>CURRENT/TIME DUPLEX</b> is a variation of duplex with current active for 0 % to 50 % output (tuning set 2) and time is active 50 % to 100 % output (tuning set 1).<br>Relay controls heat, current controls cool.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                  | TI CUR                                             | <b>ATTENTION</b> Other prompts affected: OUT RNG<br><b>TIME CURRENT DUPLEX</b> is similar to CUR TI except that current is active for 50 % to 100 % and time is active for 0 % to 50 %.<br>Relay controls cool, current controls heat.<br><b>ATTENTION</b> Other prompts affected: OUT RNG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| OUT RNG                          | 50 PCT                                             | <b>CURRENT DUPLEX RANGE ALGORITHM</b> — Used with Output Algorithm selections CUR D, CUR TI, or TI CUR.<br><b>CURRENT DUPLEX RANGE (SPLIT)</b> —This setting should be used for Relay/Current and Current/Relay Duplex Outputs. It can also be used for Current Duplex when an Auxiliary Output board is present. This enables the normal control current output to provide heat control and the auxiliary current output to provide cool control. To enable this: <ul style="list-style-type: none"> <li>AUX OUT in the Options Set Up group must be selected for Output.</li> <li>The Auxiliary Current Output is scaled as desired for 0-50 % controller output.</li> <li>Deadband for this configuration only applies to the Current Output. The Auxiliary Output must have the Deadband scaled in.</li> </ul> FOR EXAMPLE: |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display   | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                                                |         |                                             |         |                                             |         |                                          |
|----------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------|---------|---------------------------------------------|---------|---------------------------------------------|---------|------------------------------------------|
|                                  | 100PCT                                               | <p>If a 2 % Deadband is desired, then enter <b>2.0</b> for the Deadband selection in the Control Algorithm group. This will apply Deadband to the Current Output. In the Options group, set the Auxiliary Output LOW VAL selection to <b>49.0</b> and the HIGH VAL selection to <b>0.0</b>.</p> <p><b>CURRENT DUPLEX RANGE (FULL)</b> enables the Current Output to provide both heat and cool functions for control over 0-100 % of the controller output. The PID heat parameters apply when the output is greater than 50 % and the PID cool parameters apply when the output is less than 50 %. The second current output is not required for this type of duplex operation.</p> |         |                                                |         |                                             |         |                                             |         |                                          |
| <b>RLYSTATE</b>                  | 1OF 2OF<br><br>1ON 2OF<br><br>1OF 2ON<br><br>1ON 2ON | <p><b>DIGITAL OUTPUT STATUS AT 0 % OUTPUT</b> allows the following selections:</p> <table><tr><td>1OF 2OF</td><td>Output 1 de-energized<br/>Output 2 de-energized</td></tr><tr><td>1ON 2OF</td><td>Output 1 energized<br/>Output 2 de-energized</td></tr><tr><td>1OF 2ON</td><td>Output 1 de-energized<br/>Output 2 energized</td></tr><tr><td>1ON 2ON</td><td>Output 1 energized<br/>Output 2 energized</td></tr></table>                                                                                                                                                                                                                                                           | 1OF 2OF | Output 1 de-energized<br>Output 2 de-energized | 1ON 2OF | Output 1 energized<br>Output 2 de-energized | 1OF 2ON | Output 1 de-energized<br>Output 2 energized | 1ON 2ON | Output 1 energized<br>Output 2 energized |
| 1OF 2OF                          | Output 1 de-energized<br>Output 2 de-energized       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                                                |         |                                             |         |                                             |         |                                          |
| 1ON 2OF                          | Output 1 energized<br>Output 2 de-energized          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                                                |         |                                             |         |                                             |         |                                          |
| 1OF 2ON                          | Output 1 de-energized<br>Output 2 energized          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                                                |         |                                             |         |                                             |         |                                          |
| 1ON 2ON                          | Output 1 energized<br>Output 2 energized             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                                                |         |                                             |         |                                             |         |                                          |
| <b>RLY TYPE</b>                  | MECHAN<br><br>SOL ST                                 | <p><b>RELAY CYCLE TIME INCREMENT</b> selection is used only for Time Simplex and Duplex output configurations. This configuration sets the increment size of the relay cycle times in the Tuning and Tuning 2 Set Up groups.</p> <p><b>ELECTROMECHANICAL RELAY</b>—Cycle time in one-second increments.</p> <p><b>SOLID STATE RELAY</b>—Cycle time in 1/3 second increments. This is useful for solid state relay applications that require shorter cycle times. DO NOT use this setting unless cycle times of less than 1 second are required.</p> <p><b>ATTENTION</b> The Lockout selection must be set to NONE in order to view this selection.</p>                               |         |                                                |         |                                             |         |                                             |         |                                          |
| <b>MOTOR TI</b>                  | 5 to 1800 seconds                                    | <p><b>MOTOR TIME</b> – Appears only when “POSITON” is selected as the Output algorithm. This is the time it takes the motor to travel from 0 to 100% (fully closed to fully open). This time can usually be found on the nameplate of the motor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |                                                |         |                                             |         |                                             |         |                                          |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display                                                    | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CUR OUT</b>                   | DISABLE<br>INPUT 1<br>INPUT 2<br>PV<br>DEV<br>OUTPUT<br>SP<br>LSP<br>RSP<br>IN ALG 1                  | <p><b>CURRENT OUTPUT</b> – If Current Output #1 is not used to perform one of the above output algorithms, it may be used to perform an Auxiliary Output function.</p> DISABLE<br>INPUT 1<br>INPUT 2<br>PV (Process Variable)<br>DEVIATION<br>OUTPUT<br>SETPOINT<br>LOCAL SETPOINT<br>REMOTE<br>INPUT ALGORITHM 1                                                                     |
| <b>LOW VAL</b>                   | Low Scale Value within the range of the selected variable to represent the minimum output (0 or 4 mA) | <p><b>CURRENT OUTPUT LOW SCALING FACTOR</b>— Used only when CUR OUT is any selection other than DISABLE. This is a value in engineering units used to represent all CUR OUT parameters except Output.</p> <p>For Output, this is a value in percent and can be any value between –5 % and +105 %. However, keep in mind that relay output types can only be scaled 0 % to 100 %.</p>  |
| <b>HIGH VAL</b>                  | High Scale Value within the range of the selected variable to represent the maximum output (20 mA)    | <p><b>CURRENT OUTPUT HIGH SCALING FACTOR</b>— Used only when CUR OUT is any selection other than DISABLE. This is a value in engineering units used to represent all CUR OUT parameters except Output.</p> <p>For Output, this is a value in percent and can be any value between –5 % and +105 %. However, keep in mind that relay output types can only be scaled 0 % to 100 %.</p> |
| <b>CO RANGE</b>                  | 4-20mA<br>0-20mA                                                                                      | <p><b>CURRENT OUTPUT RANGE</b> allows the user to easily select 4-20 mA output or 0-20 mA output operation without the need for recalibration of the instrument.</p> <p><b>ATTENTION</b> Changing the Current Output Range will result in the loss of Field Calibration values and will restore Factory Calibration values.</p>                                                       |

## 3.9 Input 1 Set Up Group

### Introduction

This data deals with various parameters required to configure Input 1.

### Function Prompts

**Table 3-9 INPUT 1 Group Function Prompts**

| Function Prompt<br>Lower Display                                                                                                                                    | Selections or<br>Range of Setting<br>Upper Display                                                                                                                                                                                                                                                                                                                    | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IN1 TYPE</b><br><br><b>ATTENTION</b><br>Changing the input type will result in the loss of Field Calibration values and will restore Factory Calibration values. | DISABLE<br>B TC<br>E TC H<br>E TC L<br>J TC H<br>J TC M<br>J TC L<br>K TC H<br>K TC M<br>K TC L<br>NNM H<br>NNM L<br>NIC H<br>NIC L<br>R TC<br>S TC<br>T TC H<br>T TC L<br>W TC H<br>W TC L<br>100 PT<br>100 LO<br>200 PT<br>500 PT<br>RAD RH<br>RAD RI<br>0-20mA<br>4-20mA<br>0-10mV<br>0-50mV<br>0-100mV<br>0-5 V<br>1-5 V<br>0-10 V<br>TC DIFF<br>CARBON<br>OXYGEN | <b>INPUT 1 ACTUATION TYPE</b> – This selection determines what actuation you are going to use for Input 1.<br><br><b>DISABLE</b> —Disables Input.<br><b>B TC</b> —B Thermocouple<br><b>E TC H</b> —E Thermocouple High<br><b>E TC L</b> —E Thermocouple Low<br><b>J TC H</b> —J Thermocouple High<br><b>J TC M</b> —J Thermocouple Med<br><b>J TC L</b> —J Thermocouple Low<br><b>K TC H</b> —K Thermocouple High<br><b>K TC M</b> —K Thermocouple Med<br><b>K TC L</b> —K Thermocouple Low<br><b>NNM H</b> —Ni-Ni-Moly Thermocouple High<br><b>NNM L</b> —Ni-Ni-Moly Thermocouple Low<br><b>NIC H</b> —Nicrosil-Nisil Thermocouple High<br><b>NIC L</b> —Nicrosil-Nisil Thermocouple Low<br><b>R TC</b> —R Thermocouple<br><b>S TC</b> —S Thermocouple<br><b>T TC H</b> —T Thermocouple High<br><b>T TC L</b> —T Thermocouple Low<br><b>W TC H</b> —W5W26 Thermocouple High<br><b>W TC L</b> —W5W26 Thermocouple Low<br><b>100 PT</b> —100 Ohm RTD High<br><b>100 LO</b> —100 Ohm RTD Low<br><b>200 PT</b> —200 Ohm RTD<br><b>500 PT</b> —500 Ohm RTD<br><b>RAD RH</b> —Radiamatic RH<br><b>RAD RI</b> —Radiamatic RI<br><b>0-20mA</b> —0 to 20 Milliampères<br><b>4-20mA</b> —4 to 20 Milliampères<br><b>0-10mV</b> —0 to 10 Millivolts<br><b>0-50mV</b> —0 to 50 Millivolts<br><b>0-100mV</b> —0 to 100 Millivolts<br><b>0-5 V</b> —0 to 5 Volts<br><b>1-5 V</b> —1 to 5 Volts<br><b>0-10 V</b> —0 to 10 Volts<br><b>TC DIFF</b> —Thermocouple Differential<br><b>Carbon</b> —Carbon Probe Input<br><b>Oxygen</b> —Oxygen Probe Input |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display                                                                                                                                                                                                                 | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>XMITTER</b>                   | B TC      S TC<br>E TC H    T TC H<br>E TC L    T TC L<br>J TC H    W TC H<br>J TC M    W TC L<br>J TC L    100 PT<br>K TC H    100 LO<br>K TC M    200 PT<br>K TC L    500 PT<br>NNM H    RAD RH<br>NNM L    RAD RI<br>NIC H    LINEAR<br>NIC L    SQROOT<br>R TC | <p><b>TRANSMITTER CHARACTERIZATION</b>—This selection lets you instruct the controller to characterize a linear input to represent a non-linear one. If characterization is performed by the transmitter itself, then select LINEAR.</p> <p><b>ATTENTION</b> Prompt only appears when a linear actuation is selected at prompt IN1 TYPE.</p> <p>FOR EXAMPLE:<br/>If input 1 is a 4 to 20 mA signal, but the signal represents a type K H thermocouple, then configure <b>K TC H</b> and the controller will characterize the 4 to 20 mA signal so that it is treated as a type K thermocouple input (high range).</p> <p>Parameter definitions are the same as in IN1 TYPE.</p> |
| <b>IN1 HIGH</b>                  | –999. To 9999. Floating<br>(in engineering units)                                                                                                                                                                                                                  | <p><b>INPUT 1 HIGH RANGE VALUE</b> in engineering units is displayed for all inputs but can only be configured for linear or square root transmitter characterization.</p> <p>Scale the #1 input signal to the display value you want for 100 %.</p> <p>EXAMPLE:<br/>Process Variable = Flow<br/>Range of Flow = 0 to 250 Liters/Minute<br/>Actuation (Input 1) = 4 to 20 mA<br/>Characterization (XMITTER) = LINEAR<br/>Set IN1 HI display value to 250<br/>Set IN1 LO display value to 0<br/>Then 20 mA = 250 Liters/Minute<br/>and 4 mA = 0 Liters/Minute</p> <p><b>ATTENTION</b> The control setpoint will be limited by the range of units selected here.</p>              |
| <b>IN1 LOW</b>                   | –999. To 9999. Floating<br>(in engineering units)                                                                                                                                                                                                                  | <p><b>INPUT 1 LOW RANGE VALUE</b> in engineering units is displayed for all inputs but can only be configured for linear or square root transmitter characterization. Scale the #1 input signal to the display value you want for 0 %. See example above.</p> <p><b>ATTENTION</b> The control setpoint will be limited by the range of units selected here.</p>                                                                                                                                                                                                                                                                                                                 |
| <b>RATIO 1</b>                   | –20.00 to 20.00<br>Floats to 3 decimal places                                                                                                                                                                                                                      | <b>RATIO ON INPUT 1</b> —Select the Ratio value you want on Input 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>BIAS IN1</b>                  | –999. to 9999.<br>(in engineering units)                                                                                                                                                                                                                           | <b>BIAS ON INPUT 1</b> — Bias is used to compensate the input for drift of an input value due to deterioration of a sensor, or some other cause. Select the bias value you want on Input 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display      | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FILTER 1</b>                  | 0 to 120 seconds<br>No filter = 0                       | <b>FILTER FOR INPUT 1</b> —A software digital filter is provided for Input 1 to smooth the input signal. You can configure the first order lag time constant from 1 to 120 seconds. If you do not want filtering, enter 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>BURNOUT</b>                   | NONE<br><br>UP<br><br><br><br>DOWN<br><br><br><br>NO FS | <p><b>BURNOUT PROTECTION (SENSOR BREAK)</b> provides most input types with upscale or downscale protection if the input fails.</p> <p><b>NO BURNOUT</b>—Pre-configured Failsafe output (selected in the CONTROL Set up Group) applied if failed input is detected (does not apply for an input out of range). Diagnostic message IN1 FAIL is intermittently flashed on the lower display.</p> <p><b>UPSCALE BURNOUT</b> will force the Input 1 signal to the full scale value when the sensor fails. Diagnostic message IN1 FAIL intermittently flashed on the lower display.</p> <p>The controller remains in Automatic control mode and adjusts the controller output signal in response to the full scale Input 1 signal developed by the Burnout circuitry.</p> <p><b>DOWNSCALE BURNOUT</b> will force the Input 1 signal to the lower range value when the sensor fails. Diagnostic message IN1 FAIL intermittently flashed on the lower display.</p> <p>The controller remains in Automatic control mode and adjusts the controller output signal in response to the lower range Input 1 signal developed by the Burnout circuitry.</p> <p><b>NO FAILSAFE</b>— This selection does not provide input failure detection and should only be used when a thermocouple input is connected to another instrument which supplies the Burnout current. (For this selection, no burnout signal is sent to the sensor.)</p> <p><b>ATTENTION</b> For Burnout to function properly on a 0-20 mA input type (or a 0-5V type that uses a dropping resistor), the dropping resistor must be remotely located (across the transmitter terminals). Otherwise, the input at the instrument terminals will always be 0 mA (i.e., within the normal operating range) when the 0-20 mA line is opened.</p> |



| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display | Parameter<br>Definition                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMISSIV1</b>                  | 0.01 to 1.00                                       | <b>EMISSIVITY</b> is a correction factor applied to the Radiamatic input signal that is the ratio of the actual energy emitted from the target to the energy which would be emitted if the target were a perfect radiator. Available only for Radiamatic inputs. |

## 3.10 Input 2 Set Up Group

### Introduction

This data deals with various parameters required to configure Input 2.

### Function Prompts

**Table 3-10 INPUT 2 Group Function Prompts**

| Function Prompt<br>Lower Display                                                                                                                                                                                                                                                        | Selections or<br>Range of Setting<br>Upper Display                                                                                                                                                                                                                                                                                                          | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>IN2 TYPE</b><br><br><b>ATTENTION</b><br>Changing the input type will result in the loss of Field Calibration values and will restore Factory Calibration values.<br><br>Selecting Position Proportional Control in the Output Setup Group forces Input 2 to the Slidewire Selection. | DISABLE<br>B TC<br>E TC H<br>E TC L<br>J TC H<br>J TC M<br>J TC L<br>K TC H<br>K TC M<br>K TC L<br>NNM H<br>NNM L<br>NIC H<br>NIC L<br>R TC<br>S TC<br>T TC H<br>T TC L<br>W TC H<br>W TC L<br>100 PT<br>100 LO<br>200 PT<br>500 PT<br>RAD RH<br>RAD RI<br>0-20mA<br>4-20mA<br>0-10mV<br>0-50mV<br>0-100mV<br>0-5 V<br>1-5 V<br>0-10 V<br>TC DIFF<br>SLIDEW | <b>INPUT 2 ACTUATION TYPE</b> – This selection determines what actuation you are going to use for Input 2.<br><br><b>DISABLE</b> —Disables Input.<br><b>B TC</b> —B Thermocouple<br><b>E TC H</b> —E Thermocouple High<br><b>E TC L</b> —E Thermocouple Low<br><b>J TC H</b> —J Thermocouple High<br><b>J TC M</b> —J Thermocouple Med<br><b>J TC L</b> —J Thermocouple Low<br><b>K TC H</b> —K Thermocouple High<br><b>K TC M</b> —K Thermocouple Med<br><b>K TC L</b> —K Thermocouple Low<br><b>NNM H</b> —Ni-Ni-Moly Thermocouple High<br><b>NNM L</b> —Ni-Ni-Moly Thermocouple Low<br><b>NIC H</b> —Nicrosil-Nisil Thermocouple High<br><b>NIC L</b> —Nicrosil-Nisil Thermocouple Low<br><b>R TC</b> —R Thermocouple<br><b>S TC</b> —S Thermocouple<br><b>T TC H</b> —T Thermocouple High<br><b>T TC L</b> —T Thermocouple Low<br><b>W TC H</b> —W5W26 Thermocouple High<br><b>W TC L</b> —W5W26 Thermocouple Low<br><b>100 PT</b> —100 Ohm RTD High<br><b>100 LO</b> —100 Ohm RTD Low<br><b>200 PT</b> —200 Ohm RTD<br><b>500 PT</b> —500 Ohm RTD<br><b>RAD RH</b> —Radiamatic RH<br><b>RAD RI</b> —Radiamatic RI<br><b>0-20mA</b> —0 to 20 Milliamperes<br><b>4-20mA</b> —4 to 20 Milliamperes<br><b>0-10mV</b> —0 to 10 Millivolts<br><b>0-50mV</b> —0 to 50 Millivolts<br><b>0-100mV</b> —0 to 100 Millivolts<br><b>0-5 V</b> —0 to 5 Volts<br><b>1-5 V</b> —1 to 5 Volts<br><b>0-10 V</b> —0 to 10 Volts<br><b>TC DIFF</b> —Thermocouple Differential<br><b>SLIDEW</b> —Slidewire (For Position Proportional Applications) |

| Function Prompt<br>Lower Display | Selections or<br>Range of Setting<br>Upper Display                                                                                                                                                                                                                                                                       | Parameter<br>Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>XMITTER2</b>                  | B TC            S TC<br>E TC H        T TC H<br>E TC L        T TC L<br>J TC H        W TC H<br>J TC M        W TC L<br>J TC L        100 PT<br>K TC H        100 LO<br>K TC M        200 PT<br>K TC L        500 PT<br>NNM H        RAD RH<br>NNM L        RAD RI<br>NIC H        LINEAR<br>NIC L        SQROOT<br>R TC | <p><b>TRANSMITTER CHARACTERIZATION</b>—This selection lets you instruct the controller to characterize a linear input to represent a non-linear one.</p> <p><b>ATTENTION</b> Prompt only appears when a linear actuation is selected at prompt IN2 TYPE.</p> <p>FOR EXAMPLE:<br/>If input 2 is a 4 to 20 mA signal, but the signal represents a type K thermocouple, then select <b>K TC H</b> and the controller will characterize the 4 to 20 mA signal so that it is treated as a type K thermocouple input (high range).</p> <p>Parameter definitions are the same as in IN2 TYPE.</p> |
| <b>IN2 HIGH</b>                  | –999. To 9999. Floating<br>(in engineering units)                                                                                                                                                                                                                                                                        | <p><b>INPUT 2 HIGH RANGE VALUE</b> in engineering units is displayed for all inputs but can only be configured for linear or square root transmitter characterization</p> <p>See the example in IN1 HI.</p>                                                                                                                                                                                                                                                                                                                                                                                |
| <b>IN2 LOW</b>                   | –999. To 9999. Floating<br>(in engineering units)                                                                                                                                                                                                                                                                        | <p><b>INPUT 2 LOW RANGE VALUE</b> in engineering units is displayed for all inputs but can only be configured for linear or square root transmitter characterization.</p> <p>See the example in IN1 HI</p>                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>RATIO 2</b>                   | –20.00 to 20.00<br>Floats to 3 decimal places                                                                                                                                                                                                                                                                            | <b>RATIO ON INPUT 2</b> —Select the Ratio value you want on Input 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>BIAS IN2</b>                  | –999. to 9999.<br>(in engineering units)                                                                                                                                                                                                                                                                                 | <b>BIAS ON INPUT 2</b> — Bias is used to compensate the input for drift of an input value due to deterioration of a sensor, or some other cause. Select the bias value you want on Input 1.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>FILTER 2</b>                  | 0 to 120 seconds<br>No filter = 0                                                                                                                                                                                                                                                                                        | <b>FILTER FOR INPUT 2</b> —A software digital filter is provided for Input 1 to smooth the input signal. You can configure the first order lag time constant from 1 to 120 seconds. If you do not want filtering, enter 0.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>BURNOUT</b>                   | NONE                                                                                                                                                                                                                                                                                                                     | <p><b>BURNOUT PROTECTION (SENSOR BREAK)</b> provides most input types with upscale or downscale protection if the input fails.</p> <p><b>NO BURNOUT</b>—If Input 2 is being used in the Control Algorithm (such as the PV or RSP input parameter), then the pre-configured Failsafe output (selected in the CONTROL Set up Group) is applied when a failed input is detected (does not apply for an input out of range). Diagnostic message IN2 FAIL is intermittently flashed on the lower display.</p>                                                                                   |